FIN-503
Advanced derivatives
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Assignment 2 report

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Introduction

This is the second assignment of the Advanced Derivatives course. We present and answer both questions in the next sections of this report. The code related to our results is joined along this report in our submission.

Exercice 1

Consider an option to exchange the return of AMZN (Amazon stock) with the return of the SPX (S&P500) index; this payoff is

$$\left(\frac{S_T^{SPX}}{S_0^{SPX}} - \frac{S_T^{AMZN}}{S_0^{AMZN}}\right)^+$$

The current price of AMZN stock is at \$231, and the SPX index at \$6600. The dividend rate for AMZN in the last year is 0%, and for the SPX index is 1.3%. Assume a risk-free rate of r = 4.8%. The time-of-expiration of the implied volatilities provided is T = 0.288 (105 days).

After computing call prices using Black-Scholes formula, we used a gaussian copula to generate uniform samples with correlation $\rho = 0.5$. We then applied Breeden-Litzenberger results to compute the inverse CDF of the of the time-T stock value and thus transform our uniform variable into time-T stock value. Figure 1 depict the implied CDF obtained from the options data. We then computed option price and standard error using the following formulas :

$$\hat{P} = e^{-rT} \cdot \frac{1}{N} \sum_i^N \text{Payoff}_\text{i} = \mathbf{0.052997} \qquad \hat{se} = \frac{e^{-rT}}{\sqrt{N}} \cdot \sqrt{\frac{1}{N-1} \sum_i^N (\text{Payoff}_\text{i} - \hat{P})^2} = \mathbf{0.000869} \tag{1}$$

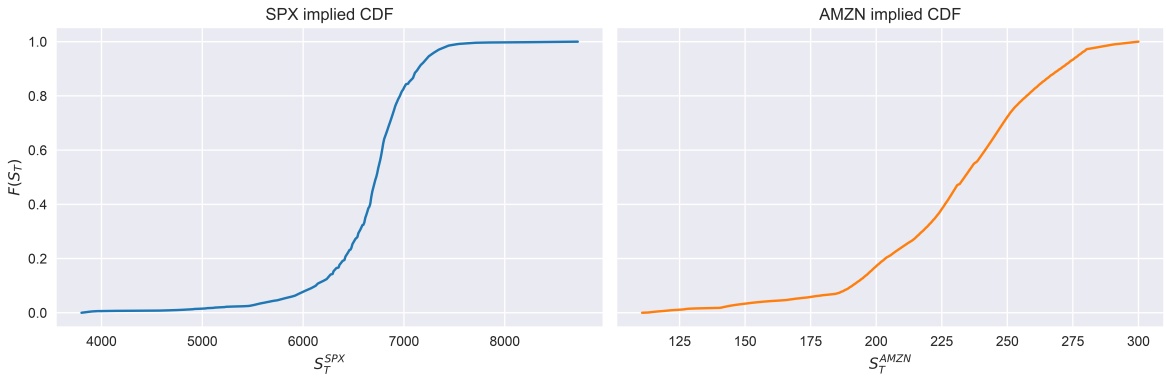


Figure 1: Implied CDF for SPX and AMZN

Exercise 2

We now price an option on AMZN stock with the same maturity $T = 0.288$, but the payoff is $\max(0, K - S^2_{AMZN}$ with $K = 218^2$. We price this option by drawing a sample of 10000 points from the uniform distribution $\mathcal{U}([0, 1])$, and we then map each point to the implied distribution for AMZN stock. By this procedure, we obtain the following option price and its standard error.

$$\hat{P} = e^{-rT} \cdot \frac{1}{N} \sum_i^N \text{Payoff}_\text{i} = \mathbf{3204.826536} \qquad \hat{se} = \frac{e^{-rT}}{\sqrt{N}} \cdot \sqrt{\frac{1}{N-1} \sum_i^N (\text{Payoff}_\text{i} - \hat{P})^2} = \mathbf{67.399098}$$

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